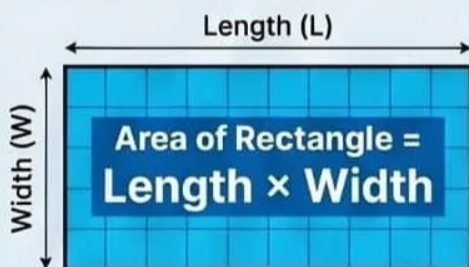


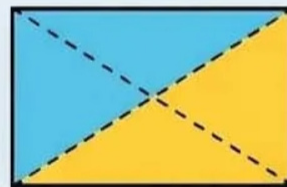
Infographic: Area



Rectangles and Squares



Area is measured by counting how many unit squares fit inside a shape.



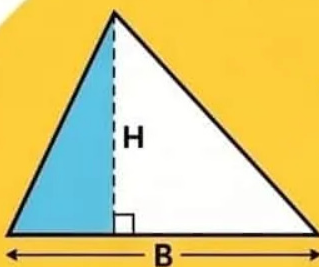
Area of a Triangle in a Rectangle:

A diagonal divides a rectangle into two congruent triangles, each having half the area.



Perimeter is not a Measure of Area: Two shapes can have the same perimeter but contain different areas.

Triangles



$$\text{Area of Triangle} = \frac{1}{2} \times \text{Base} \times \text{Height}$$



Height is the perpendicular distance from a vertex to the opposite base side.



Triangles Between Parallel Lines:

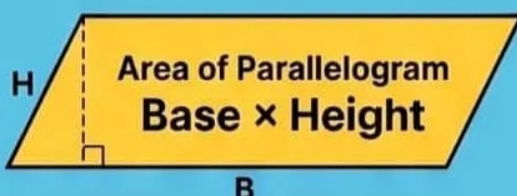
Triangles with the same base and height between parallel lines have equal areas.



Median Property:

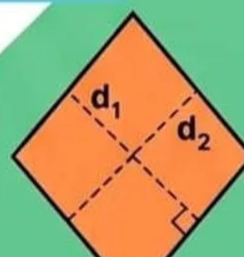
A line joining a vertex to the midpoint of the opposite side creates two equal areas.

Parallelograms



$$\text{Area of Parallelogram} = \text{Base} \times \text{Height}$$

Any side can be the base if the corresponding perpendicular height is used.

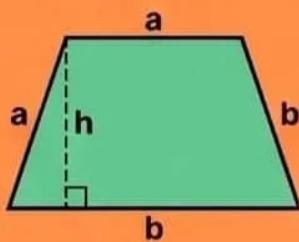


Area of Rhombus

$$= \frac{1}{2} \times d_1 \times d_2$$

Area is calculated using half the product of the two perpendicular diagonals.

Trapezium



Area of Trapezium

$$= \frac{1}{2} \times h \times (a + b)$$

Area is half the height multiplied by the sum of the parallel sides.



Area of Any Polygon

Divide the polygon into triangles, find individual areas, and add them together.

Unit Conversions

(for Real-Life Area Calculations)

Unit 1	Unit 2 (Equivalent)
1 inch	2.54 cm
1 foot	12 inches
1 acre	43,560 sq. ft